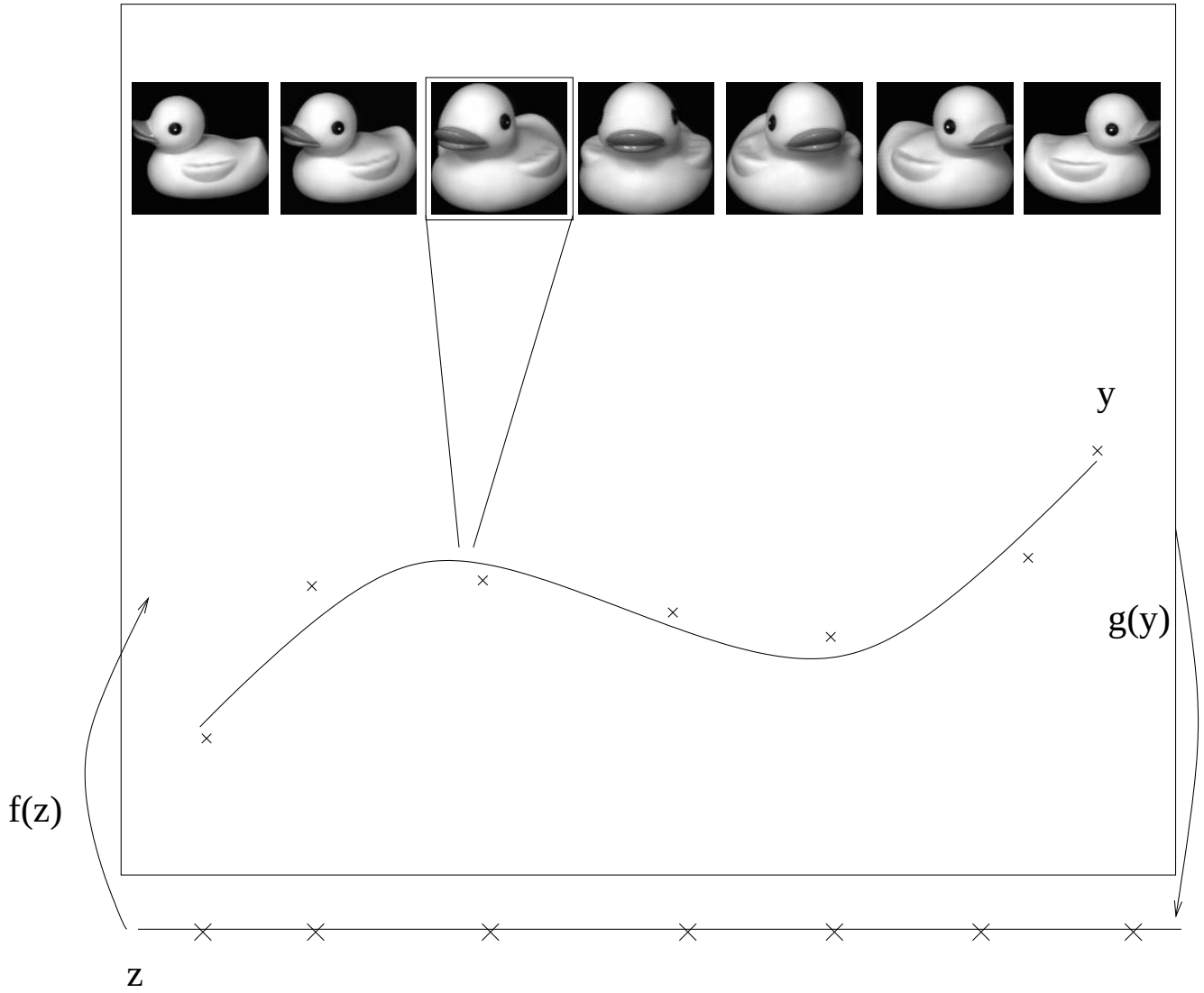
Kernel Information Embeddings

Roland Memisevic

University of Toronto

ICML 2006

Dimensionality reduction



Two issues

- Learn **mappings** f and g
  - Sammon mapping: g no, f no
  - Spectral methods (e.g. LLE, kernel PCA): g yes, f no
  - GPLVM, Principal curves: g yes, f yes (learning slow)
- Include **supervision signals**
  - Factor out things that we know already.
  - Include knowledge about grouping structure.
  - Learn the best embedding for a specific task.

Kernel Information Embedding

- An appealing criterion for embedding: **Mutual information** between y and z:

$$I(Y;Z) = \int p(y,z) \log \frac{p(y,z)}{p(y)p(z)} \, dy \, dz$$

- Intuition: "What, on average, can the low-dimensional code Z tell us about the data Y and vice versa ?"
- Can write the mutual information in terms of **entropies** $H(\cdot)$:

$$I(Y;Z) = H(Y) + H(Z) - H(Y,Z).$$

Kernel Information Embedding

- Problem: Difficult to compute in general.
- Idea: Use **kernel density estimates** to approximate entropies.
- Using $p(y,z) = \sum_j K(y,y^j)K(z,z^j)$, we can try:

$$\begin{aligned} H(Y) &= -\int p(y)\log p(y)\,dy \\ &\approx -\frac{1}{n}\sum_i \log p(y^i) \\ &\approx -\frac{1}{n}\sum_i \log \sum_j K(y^i,y^j) \\ &=: \hat{H}(Y) \end{aligned}$$

Kernel Information Embedding

- We get the MI-estimate:

$$
\begin{aligned}
& \hat{I}(Y,Z) \\
= \;& \hat{H}(Y) + \hat{H}(Z) - \hat{H}(Y,Z) \\
= \;& -\frac{1}{n}\sum_i \log \sum_j K(z^i,z^j) + \frac{1}{n}\sum_i \log \sum_j K(z^i,z^j)K(y^i,y^j) + \Omega \\
= \;& \frac{1}{n}\sum_i \log \sum_j \frac{K(z^i,z^j)}{\sum_k K(z^i,z^k)} K(y^i,y^j) + \Omega
\end{aligned}
$$

- **To train:** Gradient ascent on the z^i !

Complexity control

- Model can "cheat" by moving all z^i infinitely far apart from one another.
  - Need to restrict the range of the z^i to a finite interval. Simple way to implement this: Add variance penalty to the objective function. For example:

$$\text{maximize} \quad I(Y;Z) - \lambda \mathrm{Tr}(ZZ^{\mathrm{T}}) \quad \text{('Power constraint')}$$

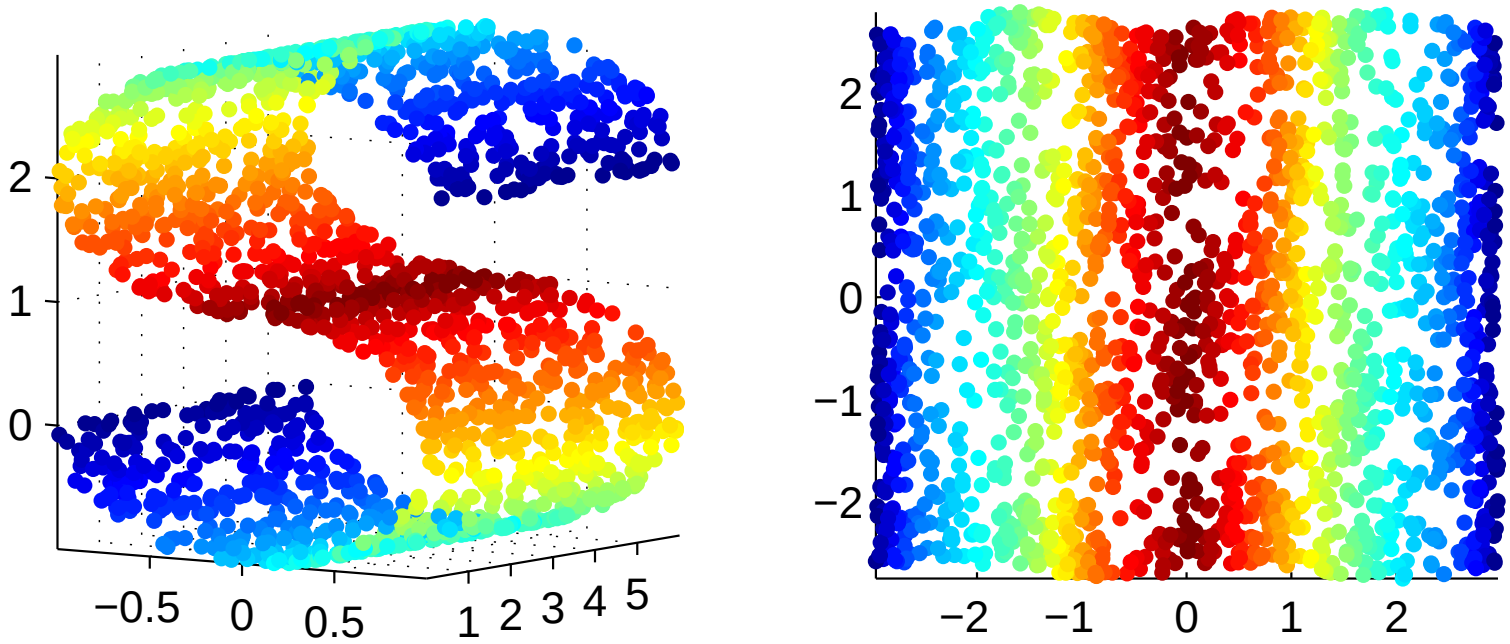
or

$$\text{maximize} \quad I(Y;Z) - \lambda \|Z\|_p$$

or ...

- Can use deterministic annealing to find good local minima: Gradually decrease λ !

Example

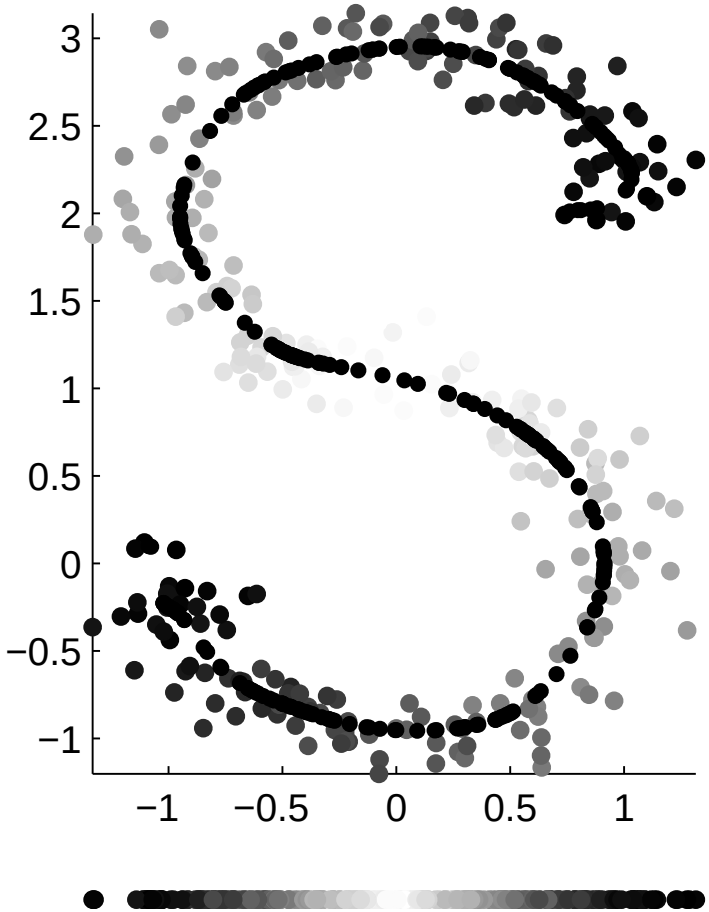


A view from non-parametric regression

$$\frac{1}{n}\sum_i \log \sum_j \frac{K(z^i, z^j)}{\sum_k K(z^i, z^k)} K(y^i, y^j) + \Omega$$

- Non-parametric regression onto latent variables [Lawrence, 2004] [Meinicke et al., 2005] [Memisevic, 2003]
- This motivates the **definitions:**
  - $f(z^{\text{test}}) = \arg\max_y \log \sum_j \frac{K(z^{\text{test}}, z^j)}{\sum_k K(z^{\text{test}}, z^k)} K(y, y^j)$
  - $g(y^{\text{test}}) = \arg\max_z \log \sum_j \frac{K(z, z^j)}{\sum_k K(z, z^k)} K(y^{\text{test}}, y^j)$

De-noising: $f(g(y))$



Conditional embeddings

- The information theoretic view suggests several ways to extend the method to **semi-supervised** settings.
- One way to introduce side-information: Replace mutual information by **conditional MI:**

$$I(Y;Z|X) \quad = \quad H(X,Z) - H(X,Y,Z) + \Omega$$

- Intuition: "What, on average, can Z tell us about Y, given that we already know X?"

Conditional embeddings

$$\textbf{maximize}: \quad \hat{I}(Y; Z|X)$$

- Capture factors Z that are not already represented by X.
- **Factor out** modes of variability that we already know about.
- In practice: Reserve some dimensions for known factors and arrange data according to those. Then **learn** the remaining, orthogonal factors.

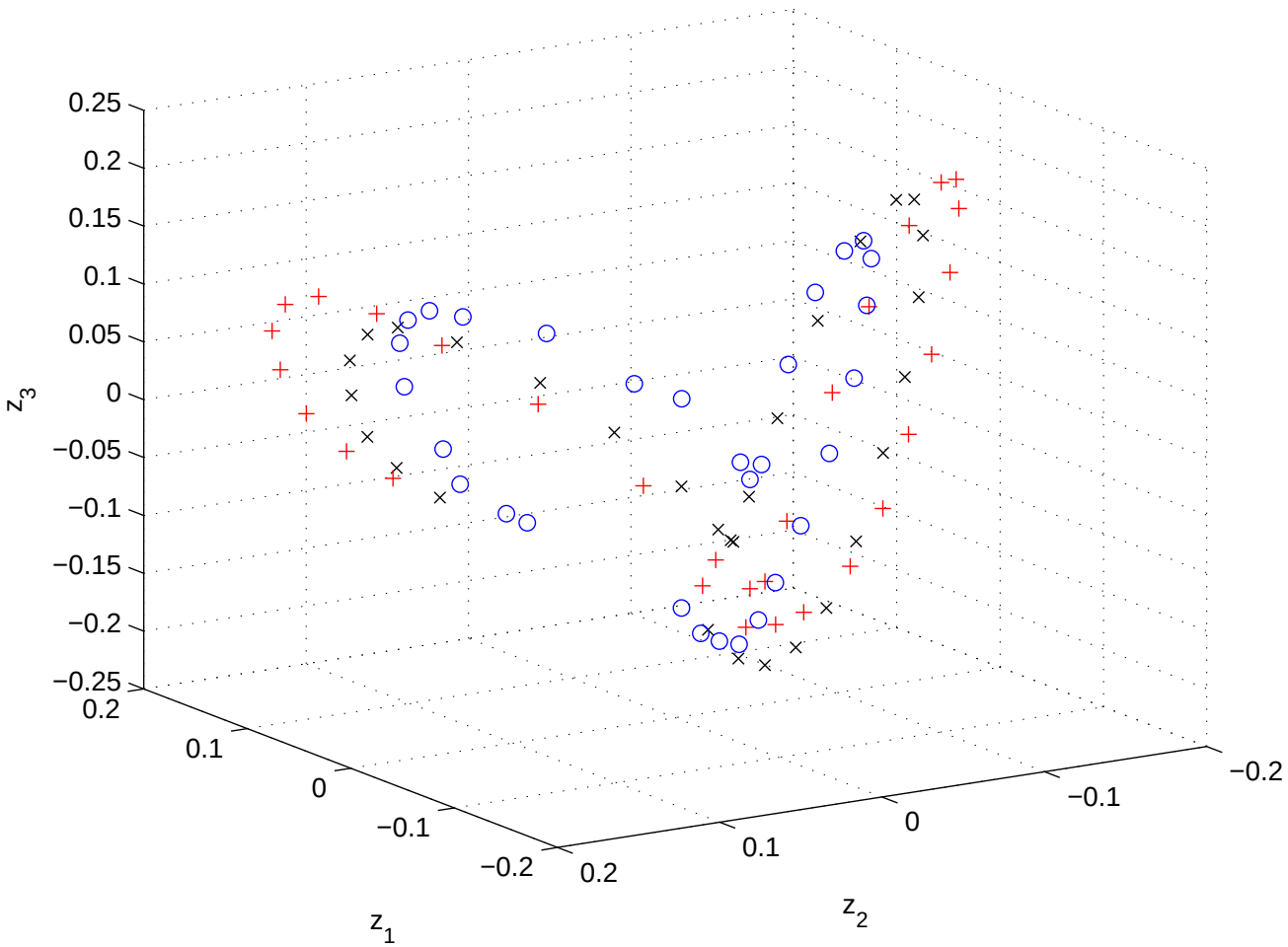
Conditional embeddings

- Example: Columbia Object Image Library (COIL)-data:



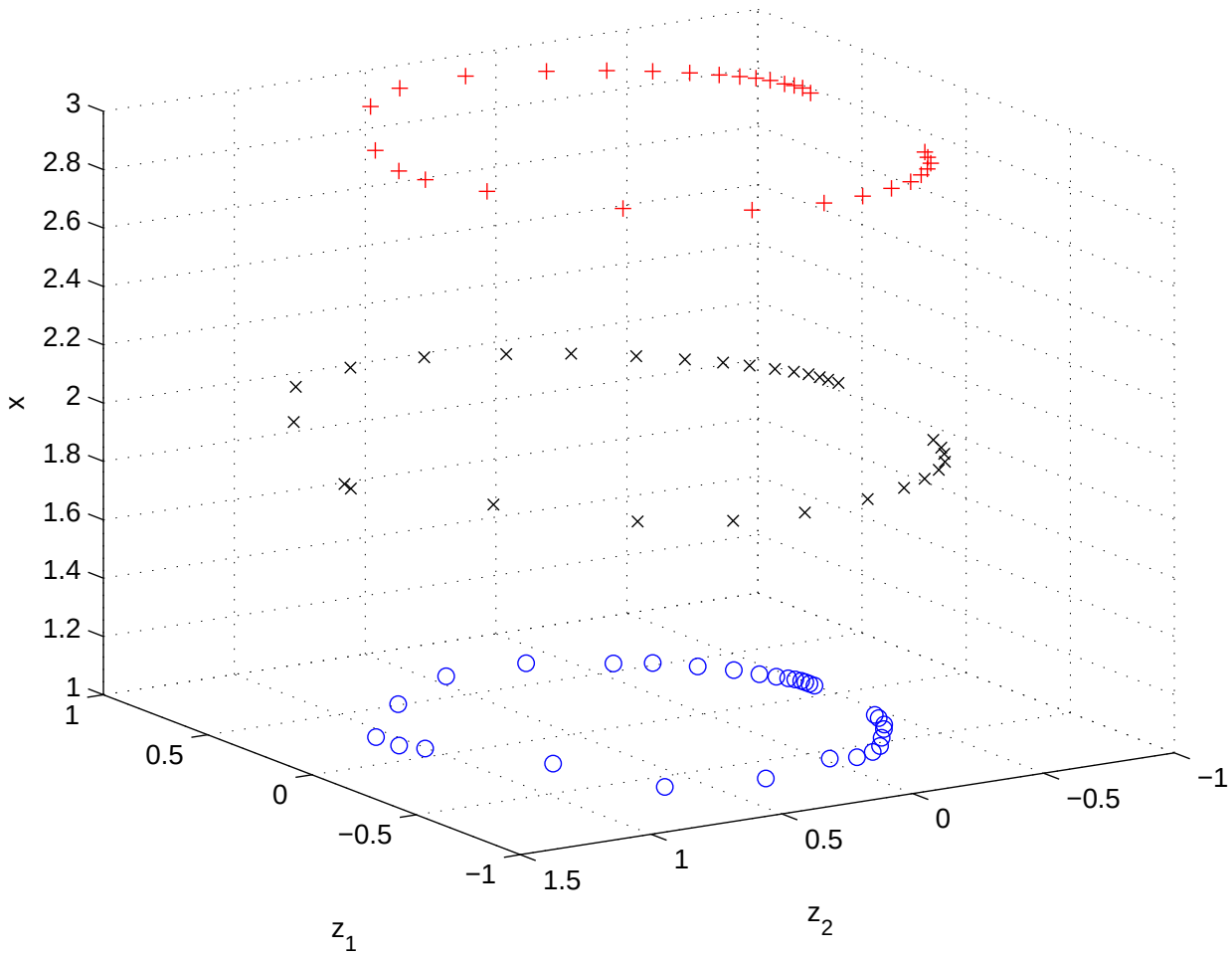
Conditional embeddings

Kernel PCA:



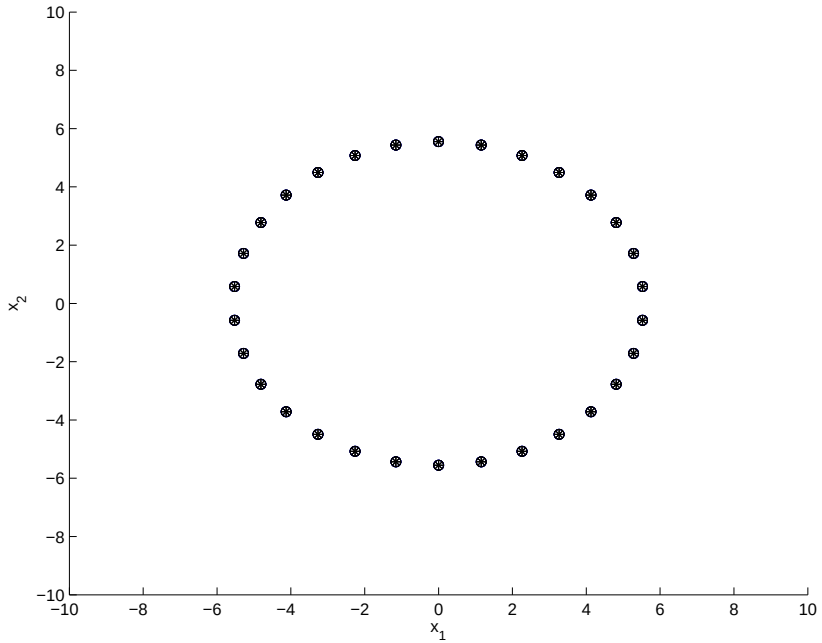
Conditional embeddings

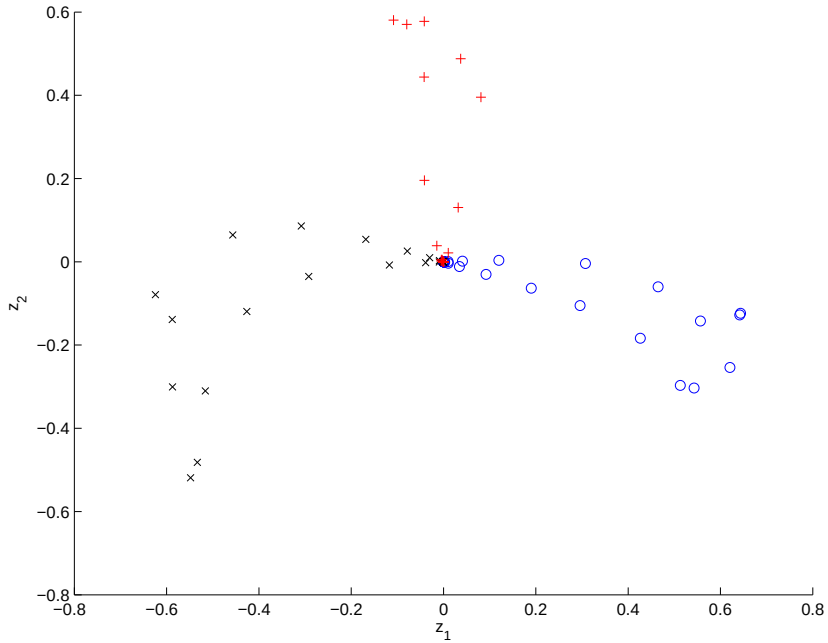
$\arg\max_Z \quad \hat{I}(Y, Z|X)$:



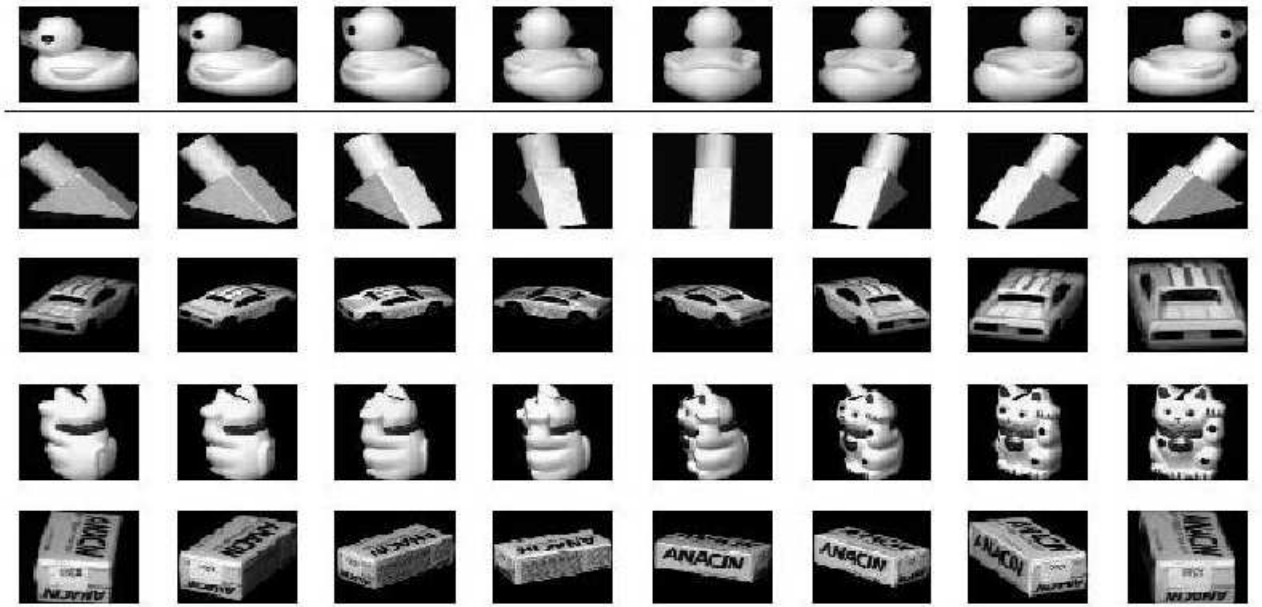
Conditional embeddings

$\arg\max_Z \quad \hat{I}(Y, Z|X)$:

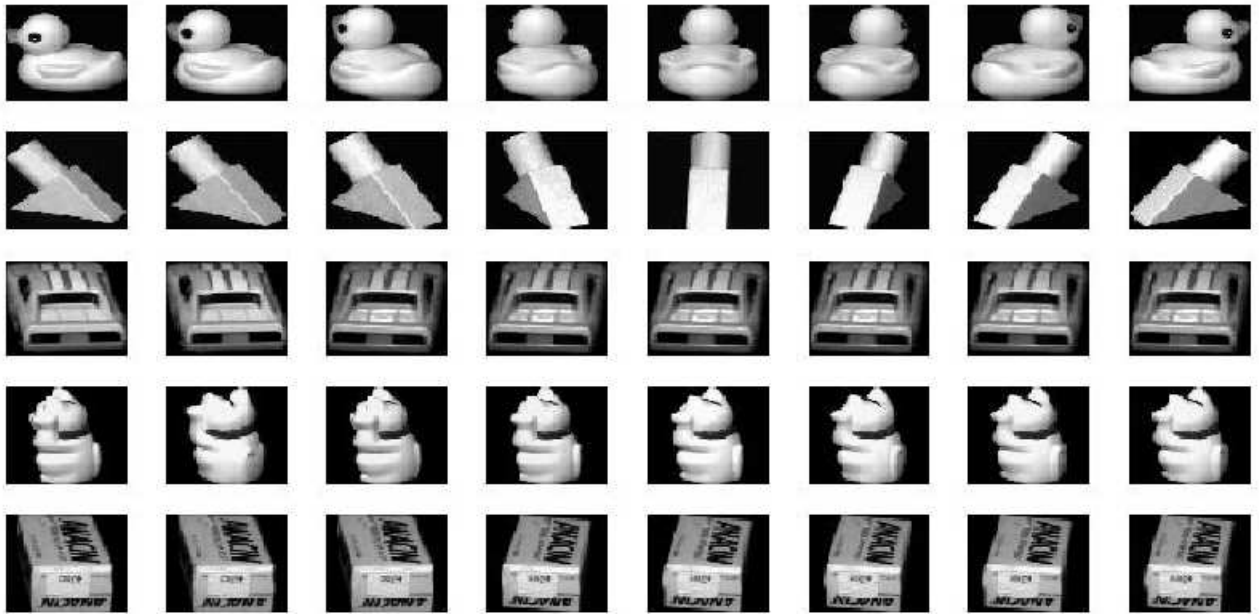




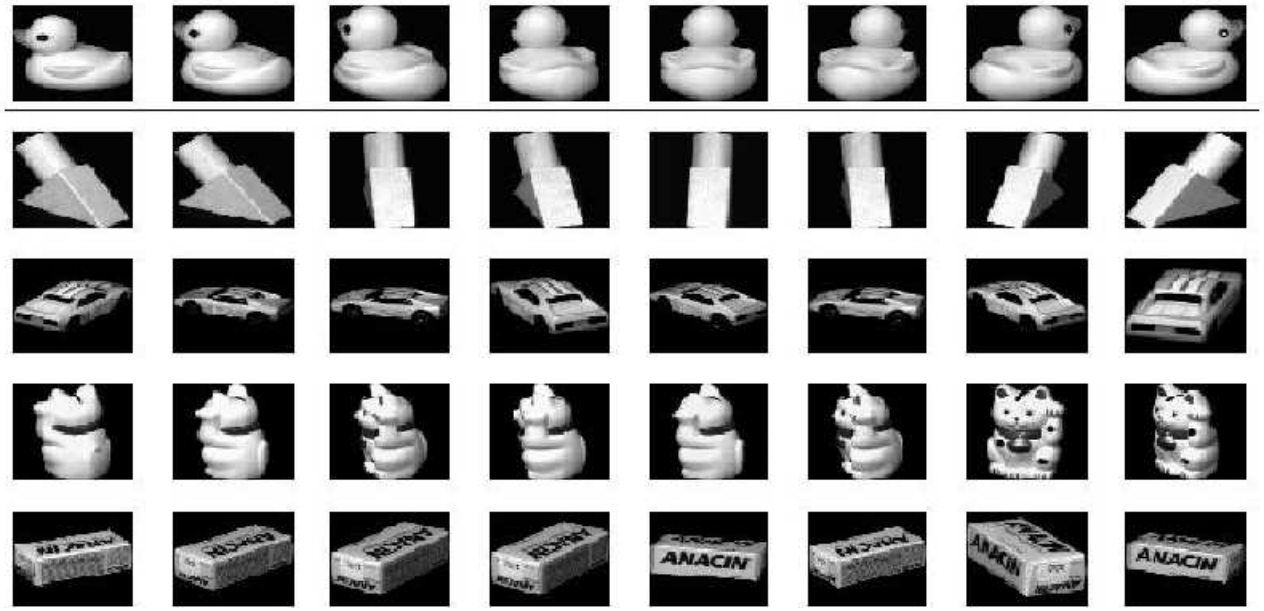
Neighborhood



Neighborhood



Neighborhood



Manifold Alignment

- Other ways to include side-information:

$$\textbf{minimize}: \quad \hat{I}(X; Y|Z)$$

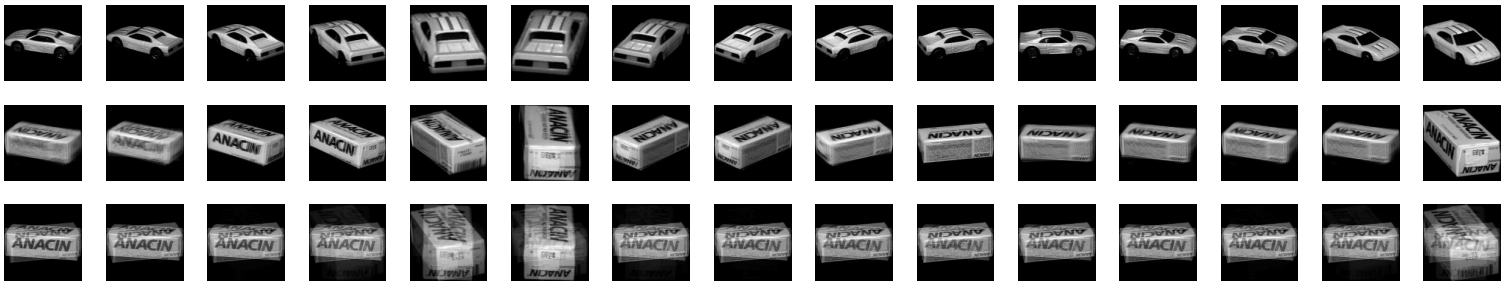
- Intuition: "Find factors Z, such that given these, X can tell us **as little** as possible about Y and vice versa."
- Capture the variability that is **shared** by X and Y: "Let $X - Z - Y$ become a Markov chain."
- Feature extraction. Manifold alignment [Ham et al., 2005].

Example: Semisupervised regression

- Training data:



- Regression result:



Conclusions

- Information theoretic embedding.
- Optimize latent representatives directly, instead of using a parametric model.
- Forward- and backward mappings available nevertheless.
- In general: embeddings need to be supplemented by supervision signal(s) to be useful.

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